

Scalability & Future Plan

Date	2 October 2026
Team ID	SWTID-2026-5625
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	-

Scalability & Future Plan:

This document outlines how FitBuddy can be scaled to handle larger user bases, increased data loads, and extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Runs only on a local deployment (Uvicorn on a local machine) and is not hosted online.	Users outside the development environment cannot access the app; no concurrent public usage.	High
2	Uses SQLite as the database, which is file-based and handles limited concurrent writes.	Slower performance and possible locking issues as the number of users and saved plans grows.	High
3	Depends on a single external service (Google Gemini API) with free-tier rate limits, and has no user login.	Plan generation may be slow or fail under heavy load; user data and plans are not tied to secure accounts.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single FastAPI instance run locally with Uvicorn.	Deploy on a cloud platform (Render / AWS / Google Cloud Run) with multiple Uvicorn workers behind a load balancer; use async endpoints for Gemini calls.
2	Data Storage	SQLite database with SQLAlchemy for user inputs and plans.	Migrate to PostgreSQL (SQLAlchemy makes this a configuration change); add regular backups and indexing on user and plan tables.
3	Performance	Every request calls the Gemini API directly to generate the 7-day workout and food plan.	Cache repeated or similar plans, use background tasks/queues for generation, and select a faster Gemini model for quick responses.
4	Security	API key stored in a .env file; basic input handling with Pydantic; no authentication.	Add user authentication (JWT/OAuth), HTTPS, rate limiting, secure secret management, and encryption of stored health information.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Core FitBuddy: personalized 7-day workout and food plans using Gemini, health-aware tips and feedback-based updates (completed, local deployment).	Completed	Working prototype that validates the idea end to end.
Phase 2	Cloud deployment, PostgreSQL migration, and user login with saved plan history.	Oct – Dec 2026	Public access, reliable storage, and a personalized experience for each user.
Phase 3	Progress tracking dashboard (weight, workouts completed), reminders, and multi-language support.	Jan – Mar 2027	Higher user engagement and wider reach.
Phase 4	Mobile app, wearable/fitness-tracker integration, and AI chat coach for daily guidance.	Apr – Aug 2027	Real-time, data-driven plan adjustments and a larger user base.